

Homework 4

*Instructor: Henry Lam***Due Date:** Oct 10, 2025

Problem 1 Let G be a cumulative distribution function and suppose that for constants $a < b$, we wish to generate a random variable X with cumulative distribution function

$$F(x) = \frac{G(x) - G(a)}{G(b) - G(a)}, \quad a \leq x \leq b.$$

(a) If random variable Y has cumulative distribution function G , then under what information is F the conditional distribution of Y ? Prove your claim.

(b) Show that an acceptance-rejection method to generate X in this case is to generate a random variable Y having cumulative distribution function G and then accept it if it lies between a and b .

Problem 2 Derive an acceptance-rejection method to generate (X, Y) with joint density function

$$f(x, y) = \frac{1}{K} \exp(-x^2 - y^2 + x \sin(xy)), \quad \forall x, y \in \mathbb{R},$$

where K is a normalizing constant. Implement your method in a computer to generate 100 pairs of (X, Y) , and plot the distributions of X and Y respectively.

Problem 3 Derive a procedure to sample uniformly from the region

$$A = \left\{ (x, y) : x \in [-1, 1], |y| \leq |x|^{-1/2} \right\}.$$

Implement your procedure in a computer to generate 100 pairs of (X, Y) , and plot their positions on a two-dimensional plane to check that they indeed appear uniformly distributed over the region A .

Problem 4 Consider the discrete random variable X where

$$\mathbb{P}(X = j) = \left(\frac{1}{2}\right)^{j+1} + \left(\frac{1}{2}\right) \frac{2^{j-1}}{3^j}, \quad j = 1, 2, \dots$$

which appeared in HW2 Problem 3. Instead of using the inverse transform method as there, now present a composition method to generate X . Then implement your procedure in a computer to generate 100 copies of X , and plot their distribution.

Problem 5 Present a method to generate a random variable X having cumulative distribution function

$$F(x) = \int_0^\infty x^y e^{-y} dy, \quad 0 \leq x \leq 1.$$

Implement your procedure in a computer to generate 100 copies of X , and plot their distribution.